Roll Number:_____

Thapar Institute of Engg & Technology, Patiala

Department of Computer Science & Engineering

(Auxiliary Examination)

B. E. -COE	**Course Code: UCS310**
Date: 16 Feb 2025 | 10:00 AM	**Course Name: Database Management System**
Time: 3 Hours, M. Marks: 100	**Name of Faculty: Dr. Sanjeev Rao (SRO)**

Note: Make suitable assumption(s), if any.

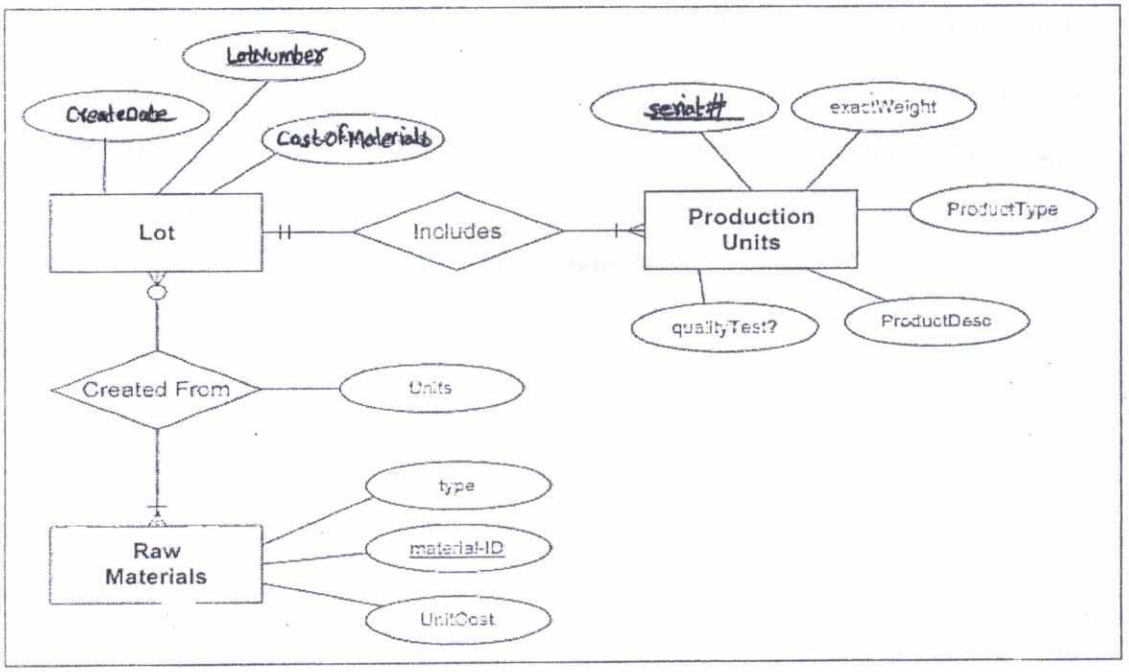
Q1 (a). Consider a FD {A-> BC, B-> AC, C-> AB}. Find canonical cover for given FD. Also, give appropriate reasoning at each step.

Q1 (b). Consider R = {A, B, C, D, E, F, G, H} with a set of FDs

$$F = \{CD \rightarrow A, EC \rightarrow H, GHB \rightarrow AB, C \rightarrow D, EG \rightarrow A, H \rightarrow B, BE \rightarrow CD, EC \rightarrow B\}$$

Find all the candidate keys of R. Give appropriate reasoning at each step. **(10*2=20)**

Q 2. Production tracking is important in many manufacturing environments (e.g., the pharmaceuticals industry, children's toys, etc.). The following ER diagram captures important information in the tracking of production. Specifically, the ER diagram captures relationships between production lots (or batches), individual production units, and raw materials.



Convert the ER diagram into a relational database schema. Indicate strong entities with attributes primary keys, foreign keys, cardinalities (Type of relationships) and referential integrity constraints with appropriate reasoning. **(20)**

Q 3. Create a trigger to ensure that the job of any new employee being hired to department 40 (the IT department) is IT_MGR or IT_PROG. Create a trigger called CHK_Trig_JOB. Fire the trigger before every row that is changed after insertions and updates to the JOB_ID column in the EMPLOYEES table. Check that the new employee has a job ID of IT_MGR or IT_PROG in the EMPLOYEES table. Add exception handling and provide an appropriate message so that the update fails if the new job ID is not IT_MGR or IT_PROG. **(20)**

Q 4). Write the SQL queries for following questions:

(a) Find the names of departments where at least one employee earns more than the overall average salary across all employees.

(b) List each employee's first name, last name, and the first name and last name of their manager.

(c) Write a SQL subquery using given employee schema to display the department number (dept_id) and the lowest salary of the department with the highest average salary.

(d) Display total number of employee working in each job in each dept. **(5*4=20)**

Q 5. Explain the following with suitable examples:

(a). Two-phase locking protocol

(b). Concurrency control mechanism

(c). 3NF and BCNF

(d). Generalization and Specialization **(5*4=20)**

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